

Inferring Collective Diffusion Regimes from Partial Agent Observation

Project Report: Phases 1–3

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Abstract

This report documents three completed phases of a project that asks whether the diffusion regime of a whole population of agents can be inferred from a small, partially observed subset of them, using each observed agent’s own trajectory history together with the instantaneous mean field it experiences. Phase 1 builds and validates a toolbox of seven generative mechanisms and their estimators and shows that a single trajectory can fail, in principle, to reveal the population’s exponent. Phase 2 quantifies exactly how estimators behave as a function of trajectory length and turns mechanism identification into a practical classification and bias-correction pipeline. Phase 3 couples agents through a long-memory mean field and shows that coupling destroys a superdiffusive agent’s own signature but not a subdiffusive one, and separates what a single agent can learn from what only the collective field reveals. Phase 4 (heterogeneous sub-populations) has not been run; Phase 3’s results are its precondition.

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1 Introduction

1.1 Research question

Given a population of agents whose microscopic motion rule is unknown, and given access to only a subset of them, can the **diffusion regime of the whole population** (subdiffusive, normal, or superdiffusive) be identified from (i) an observed agent’s own trajectory history and (ii) the instantaneous mean field it experiences? The project attacks this question through four sub-questions, in the order they are addressed:

Q1 — Estimator quality.

For each generative mechanism, how well can the anomalous exponent be recovered from a single finite trajectory? Where does a single trajectory fail *in principle* (ergodicity breaking) rather than merely from finite-sample noise?

Q2 — Mechanism discrimination.

Which observables separate mechanisms that share the same exponent (e.g. fractional Brownian motion vs. continuous-time random walks vs. Lévy walks)?

Q3 — Single agent vs. collective.

Once agents are coupled through a mean field, how correlated is a single agent’s inferred regime with the ensemble’s true regime? Does adding the mean-field signal as an extra observable help?

Q4 — Sampling cost under heterogeneity.

For a population made of M modules with different exponents, how many observed agents n are needed to recover the population’s exponent mixture, and how does the threshold ($\sim M$? $\sim M \log M$?) depend on the sampling strategy?

Phases 1–3 of this report answer Q1–Q3. Q4 is Phase 4 and has not been run; Section 5 states what Phase 3 established as its precondition.

1.2 Background: anomalous diffusion and ergodicity breaking

For an ensemble of independent trajectories, the ensemble-averaged mean squared displacement scales as a power law in time, $\langle x^2(t) \rangle \sim t^\alpha$. $\alpha = 1$ is ordinary (Brownian) diffusion; $\alpha < 1$ is *subdiffusion* (the population spreads slower than a random walk, typically because of trapping or negatively correlated steps); $\alpha > 1$ is *superdiffusion* (the population spreads faster, typically because of long, persistent excursions). α is the single number this project spends most of its effort trying to recover from limited data.

A subtlety drives half of this report: for some mechanisms, a single, arbitrarily long trajectory does not reveal α , even in principle. This is **ergodicity breaking**. An ergodic process’s *time-averaged* mean squared displacement (built from one trajectory, sliding a lag window along it) converges to the same exponent as the *ensemble-averaged* one (built from many trajectories at fixed time) as the trajectory gets long. A non-ergodic process’s time-average does not converge to the ensemble exponent — it may converge to a different number entirely, or it may keep a persistent, non-vanishing spread across realizations no matter how long the trajectory runs. Whether a mechanism is ergodic, and what its time-average does instead, is mechanism-specific and is one of the central facts Phase 1 establishes empirically (Table 1).

1.3 Conventions that govern every result in this report

Four rules, fixed before Phase 1 began, are load-bearing for everything that follows and are stated here so later sections can rely on them without re-deriving them:

- **Validation first.** A generator is not used for analysis until a test asserts that it reproduces its analytic ensemble MSD within a stated tolerance, over a stated time window. No result in this report comes from an unvalidated generator.
- **Determinism.** Every stochastic function takes an explicit random number generator; there are no calls to hidden global random state. Every figure and every number in this report is reproducible from its configuration and seed.
- **Fitting windows are explicit and asymmetric between two cases.** Power-law exponents are fitted by least squares on $\log(\text{lag})$ or $\log(t)$ versus $\log(\text{MSD})$, but *time-averaged* MSD versus lag and *ensemble-averaged* MSD versus time are not the same statistical claim and use different windows: the former is restricted to $[\text{lag}_{\min}, T/10]$ because the number of independent lag windows collapses beyond that; the latter’s window is instead an assertion about where the asymptotics of a specific mechanism hold, and is stated per case.
- **Uncertainty comes from resampling particles, never from fit residuals.** Ensemble-averaged MSD points at different times are built from the same particles and are strongly correlated, so ordinary least-squares standard errors are not a valid confidence interval (Section 2.4 quantifies this by how much). Every reported uncertainty in this document is a bootstrap over particles or an empirical seed-to-seed spread.

2 Phase 1 — Single-agent and ensemble diffusion, no interaction

2.1 Problem

Phase 1 answers the foundational half of Q1 and Q2 without any coupling between agents: build a library of generative mechanisms that covers the qualitatively different ways a particle can diffuse anomalously, validate each one against its analytic ensemble behaviour, build the estimator toolbox needed to characterize a trajectory, and use both to answer, empirically: *how well does a single trajectory reveal the true exponent, and does that answer depend on which mechanism generated the data?*

2.2 Mechanisms studied

Table 1 lists the seven mechanisms implemented and validated in Phase 1, together with the properties that distinguish them. Two entries carry the most weight for the rest of the project. **CTRW** (continuous-time random walk) is weakly ergodicity-breaking: its time-averaged MSD grows *linearly* in the lag regardless of the true exponent a , and the amplitude of that linear growth is itself a random variable that does not converge as the trajectory lengthens — a single trajectory cannot reveal a no matter how long it runs. Its jump lengths are Gaussian (finite variance), a deliberate choice: the point of contrasting CTRW against fBm at $H < 1/2$ is that they share an exponent but not an ergodic property, and heavy-tailed jumps would blur that contrast into the Lévy walk. **Lévy flight** has a diverging second moment, so MSD itself is not a valid statistic for it; any code path that would compute one is made to raise rather than silently return a finite number.

Table 1: Generative mechanisms and their analytic signatures. α is the ensemble MSD exponent, $\langle x^2(t) \rangle \sim t^\alpha$.

Mechanism	Regime	α	Ergodic	Gaussian prop.	Increment ACF
Brownian	normal	1	yes	yes	delta
fBm, $H < 1/2$	sub	$2H$	yes	yes	negative
fBm, $H > 1/2$	super	$2H$	yes	yes	positive
CTRW, $a \in (0, 1)$	sub	a	no	no	zero + long pauses
Lévy walk, $\gamma \in (1, 2)$	super	$3 - \gamma$	no	no	ballistic stretches
Lévy flight, $s \in (0, 2)$	super	diverges	no	no (stable law)	delta
Scaled Brownian, a	either	a	no	yes	delta, non-stationary
Diffusing diffusivity	normal	1	yes	no (exp. tails)	delta

2.3 Approach

The pipeline is generators $\rightarrow$ validation $\rightarrow$ estimator toolbox $\rightarrow$ visualization, strictly in that order, because validation gates everything downstream: no analysis is ever run on a generator that has not been checked against its own analytic ensemble MSD. The estimator toolbox implements time-averaged and ensemble-averaged MSD with power-law fitting, the ergodicity-breaking parameter $EB(\Delta) = \text{Var}(\xi)$ with $\xi = \delta^2 / \langle \delta^2 \rangle$, the van Hove self-part (propagator shape), the increment autocorrelation, and the moment scaling spectrum $\nu(q)$ (fit $\langle |x|^q \rangle \sim t^{q\nu(q)}$; a piecewise-linear $\nu(q)$ with a kink is the Lévy walk’s fingerprint of strong anomalous diffusion).

2.4 Bottlenecks and how they were resolved

- **Fit-residual error bars underestimate the true uncertainty by roughly $25\times$.** The ensemble MSD is built from the same particles at every time point, so its points are strongly autocorrelated and ordinary least-squares standard errors are far too optimistic: measured over twelve seeds ($N=2000$, $T=1024$), the reported fit error was ~ 0.0005 while the actual seed-to-seed standard deviation of the fitted exponent was ~ 0.012 . This is why every uncertainty in this report is a bootstrap over particles, never `PowerLawFit.exponent_err` (documented as a goodness-of-fit diagnostic only).
- **CTRW’s finite-time exponent bias changes sign with a .** The mean renewal count has a finite-time correction whose two terms pull in opposite directions: an additive -1 term biases

the fitted exponent *upward* at small a (few renewals), while a slowly-decaying $(\tau_0/t)^{1-a}$ term biases it *downward* as $a \rightarrow 1$. The measured bias at $T = 4096$ runs $+0.069$ at $a = 0.3$ down to -0.071 at $a = 0.9$, crossing zero near $a \approx 0.6$ (later refined to $a = 0.604$ at $T=4096$, drifting to 0.566 by $T=65536$ — the crossing itself is T -dependent). Consequence: a naive “bias shrinks as T grows” test would be wrong, since the bias is a property of the observation window, not of the estimator alone.

- **A log-space Jensen bias, found on review, was masquerading as part of the CTRW bias above.** A reviewer’s independently-derived bias estimate ($+0.046$ at $a = 0.3$) disagreed with the original ($+0.069$). The extra $+0.023$ turned out to be a genuine, separate effect: fitting $\log(\widehat{\text{MSD}})$ introduces a bias of $-\sigma_{\text{rel}}^2/2$ per Jensen’s inequality, and because the CTRW’s relative variance *falls* with t at small a , this tilts the log-log fit upward at finite ensemble size N . Measured at $a = 0.3$, $T = 4096$: bias $+0.067$ at $N=2000$, falling to $+0.042$ at $N=20000$, converging on the closed-form prediction of $+0.043$. This bias affects any log-log fit whose relative variance varies with the fitting abscissa, and bootstrapping gives a correct confidence interval but does not remove it.
- **The Lévy walk’s MSD estimator gets noisier, not more precise, with time.** Because the fourth moment grows as $t^{5-\gamma}$, faster than the square of the second moment ($t^{6-2\gamma}$), the relative standard error of the ensemble MSD grows roughly as $t^{(\gamma-1)/2}$: measured at $\gamma = 1.4$, $N = 3000$, it runs from 1.9% at $t = 10$ to 5.7% at $t = 4095$. This is an independent reason (on top of the window-count argument) to cap the TA-MSD fitting window.
- **Generators, not estimators, dominate memory.** At $N=4000$, $T=4096$ (one `float64` trajectory array is 125 MiB), the Lévy walk peaks at 1324 MiB and fBm at 937 MiB — roughly $10\times$ and $7.5\times$ the raw output array, from the event arrays and the complex FFT buffers respectively (Table 2). Phase 2’s sweeps generate in particle chunks rather than raising `n_particles` directly.

Table 2: Representative per-operation cost, $N=4000$, $T=4096$ (`results/benchmark.table.json`).

Operation	Time (ms)	Peak memory (MiB)
generate Brownian	174	266
generate fBm	524	937
generate SBM	165	266
generate CTRW	1085	267
generate Lévy walk	2157	1324
generate Lévy flight	820	750
generate diffusing diffusivity	561	641
TA-MSD (32 lags)	1296	376
EB curve (32 lags)	1295	376
moment spectrum (16 q)	2234	250
fit_powerlaw (200 bootstrap)	208	137

2.5 Final result: the single-trajectory exponent table

Table 3 is Phase 1’s closing measurement: 200 independent trajectories per mechanism per length, each fitted independently on its own time-averaged MSD, at four lengths spanning a 64-fold range.

The column p is the fitted exponent of $\text{IQR} \sim T^{-p}$: a larger p means the trajectory-to-trajectory spread shrinks faster as T grows.

Table 3: Single-trajectory TA-MSD exponent estimate, 200 trajectories per cell (`results/exponent_spread.json`). p is the fitted IQR shrinkage exponent, $\text{IQR} \sim T^{-p}$.

Mechanism	True α	Median $\hat{\alpha}$ ($T=$)				p
		128	512	2048	8192	
Brownian	1.00	0.958	0.976	0.997	0.990	0.263
fBm, $H = 0.3$	0.60	0.599	0.585	0.590	0.595	0.283
Scaled Brownian, $a = 0.6$	0.60	0.930	0.948	0.974	0.976	0.260
CTRW, $a = 0.5$	0.50	0.926	0.916	0.860	0.881	0.107
Diffusing diffusivity	1.00	0.966	0.972	0.987	0.991	0.264
Lévy walk, $\gamma = 1.5$	1.50	1.622	1.522	1.489	1.472	0.050

Two qualitatively different failure modes sit inside this table, and the distinction matters for Phase 2:

- **Scaled Brownian motion converges as tightly as Brownian motion, onto the wrong number.** Its IQR shrinks at essentially the ergodic rate ($p=0.260$, indistinguishable from Brownian’s $p=0.263$), while its median walks toward 1.0 against a true exponent of 0.6. A single agent’s own error bar gives no warning here: the estimate is precise and wrong.
- **CTRW fails in the other direction:** the median is wrong *and* the spread barely shrinks ($p=0.107$, roughly a third of the ergodic rate) — both are the same non-ergodicity, showing up as bias in one statistic and as a stalled interquartile range in the other. CTRW and scaled Brownian motion both report $\hat{\alpha} \approx 1$ from one trajectory, for unrelated reasons (weak ergodicity breaking vs. ageing); that coincidence is the concrete form of the Q1 information wall, and it is exactly what makes single-trajectory mechanism classification hard — the problem Phase 2 exists to address.

A second, independent check of the same non-ergodicity: for CTRW at $a=0.5$, $N=2000$, $T=4096$, the measured ergodicity-breaking parameter at lags 10/50/100 is 0.700/0.866/1.009 against a theoretical plateau of $\pi/2 - 1 = 0.571$ — systematically above the asymptotic plateau because EB rises with lag/T and converges to the plateau only slowly from above (Table 4, Figure 2). Brownian motion’s own EB, measured at the same lags, matches its closed form $4\Delta/(3T)$ to within 5–9% and serves as the positive control: a plateau scaled by the wrong constant would still look like a plateau, but this formula has a specific slope *and* coefficient, and it is met.

2.6 Figures

Table 4: Ergodicity-breaking parameter checks, $N=2000$, $T=4096$ (`results/ergodicity_checks.json`).

	lag			target
	10	50	100	
Brownian EB, measured	0.00354	0.01734	0.03442	
Brownian EB, predicted $4\Delta/(3T)$	0.00326	0.01628	0.03256	
CTRW ($a=0.5$) EB, measured	0.700	0.866	1.009	plateau 0.571

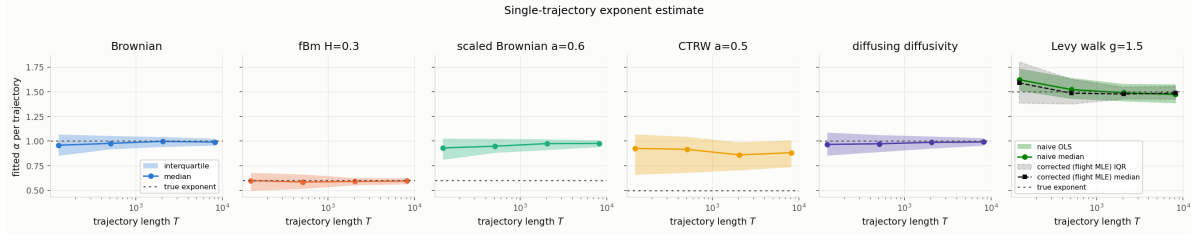


Figure 1: Single-trajectory exponent estimate vs. trajectory length T , median and interquartile band, for six mechanisms (`results/exponent_spread.json`). Read off: Brownian, fBm and diffusing diffusivity converge tightly onto their true exponent (dashed line); scaled Brownian motion converges just as tightly but onto the *wrong* number; CTRW’s band barely narrows and sits off-target; the Lévy walk panel overlays the naive OLS estimate against the flight-duration-MLE correction from Phase 2 (Section 3.3), showing the correction pulling the band down onto the true exponent.

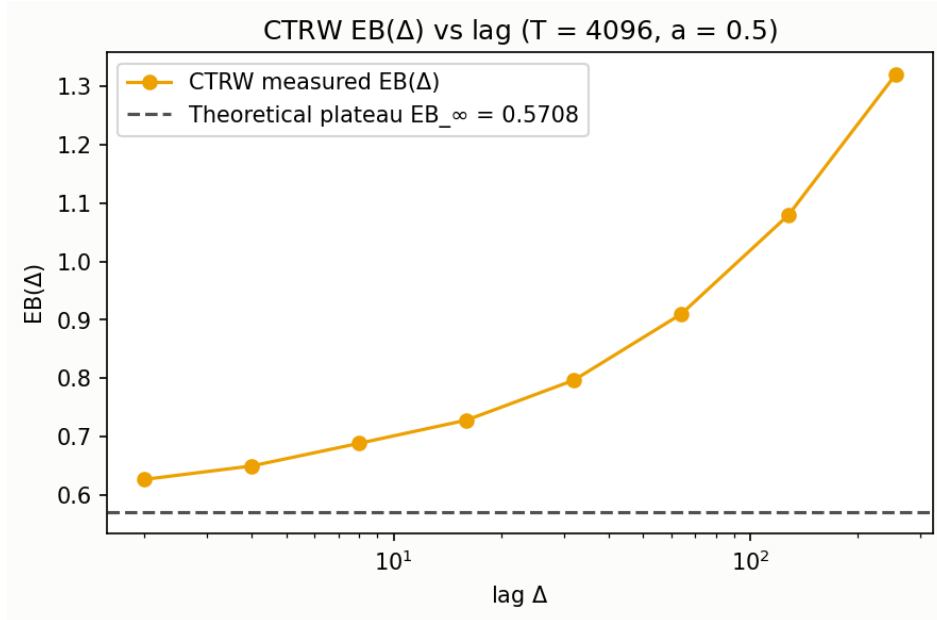


Figure 2: CTRW ergodicity-breaking parameter $EB(\Delta)$ vs. lag, $a=0.5$, $T=4096$. The curve sits above the theoretical plateau ($\pi/2 - 1 = 0.571$, dashed) and keeps rising with lag/T rather than flattening within the plotted range — the plateau is approached slowly from above, not reached at any lag shown here.

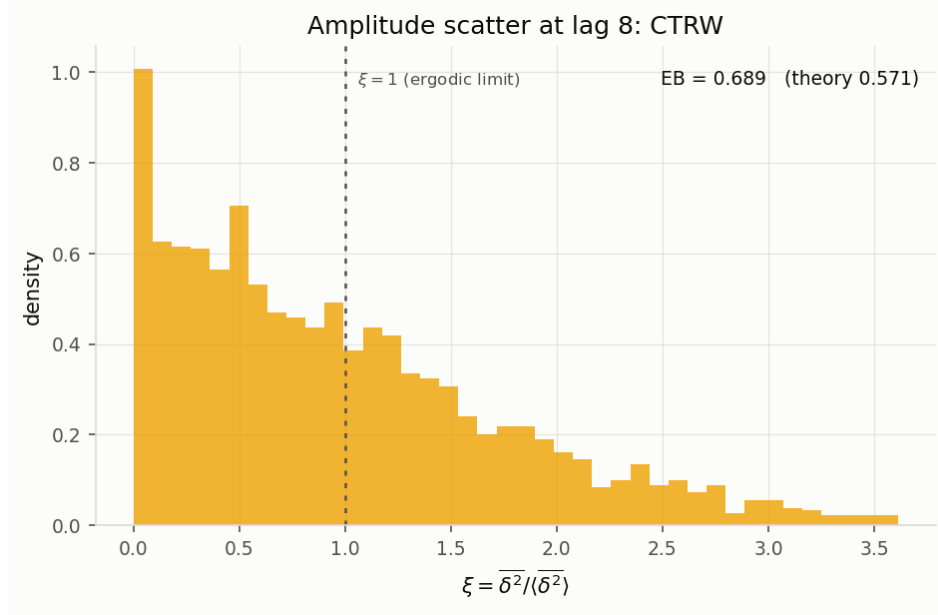


Figure 3: Distribution of the normalized time-averaged-MSD amplitude $\xi = \overline{\delta^2} / \langle \delta^2 \rangle$ at lag 8 across 2000 independent CTRW trajectories ($a=0.5$). An ergodic process would concentrate this distribution at $\xi=1$; instead it has a finite density at $\xi \rightarrow 0$ (trajectories that spent the whole measurement inside one long trapping event) and a heavy right tail. The measured EB=0.689 sits close to its theoretical value 0.571 for this one realization.

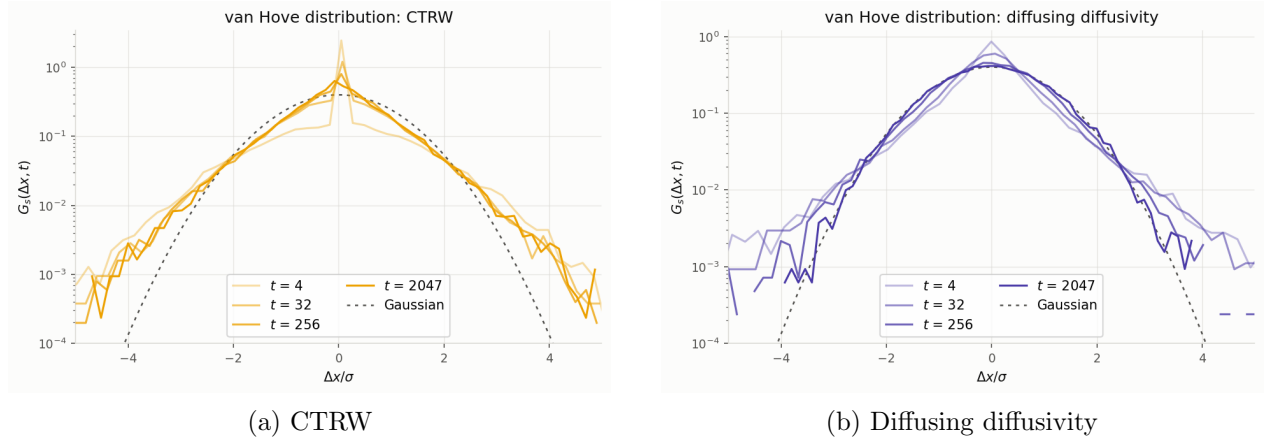


Figure 4: Self-part of the van Hove distribution $G_s(\Delta x, t)$ at several times, versus a Gaussian reference (dashed). CTRW develops a sharp central peak and non-Gaussian tails driven by particles still trapped in their first waiting interval; diffusing diffusivity keeps a symmetric but sub-Gaussian central peak with exponential (not Gaussian) tails at short times, the signature of a fluctuating local diffusion coefficient. Both mechanisms have $\alpha = 1$ at the level of the MSD alone — the propagator shape is what separates them, motivating the feature set Phase 2 builds on.

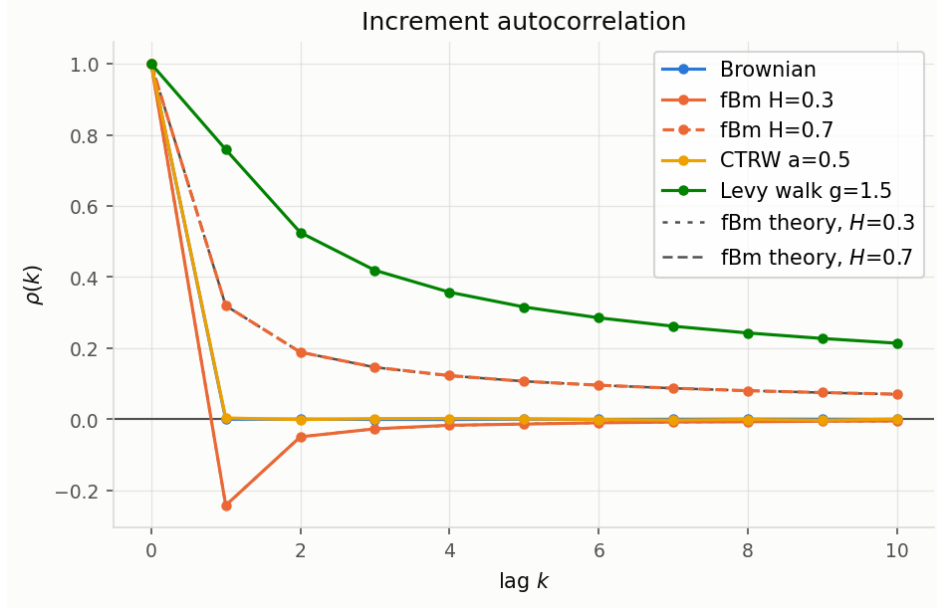
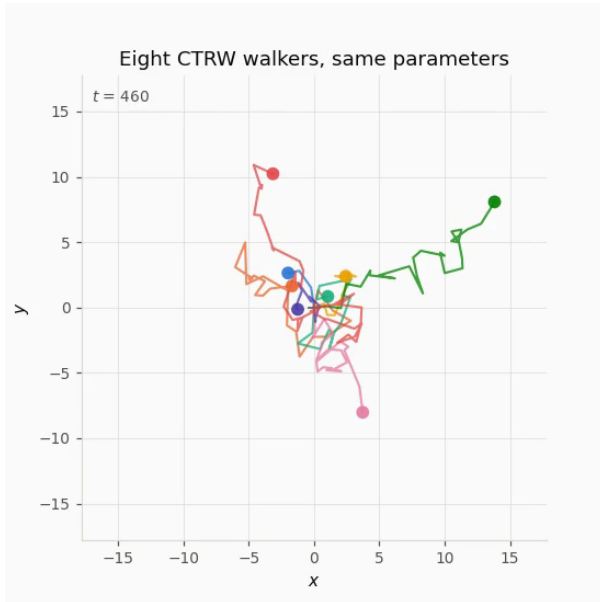
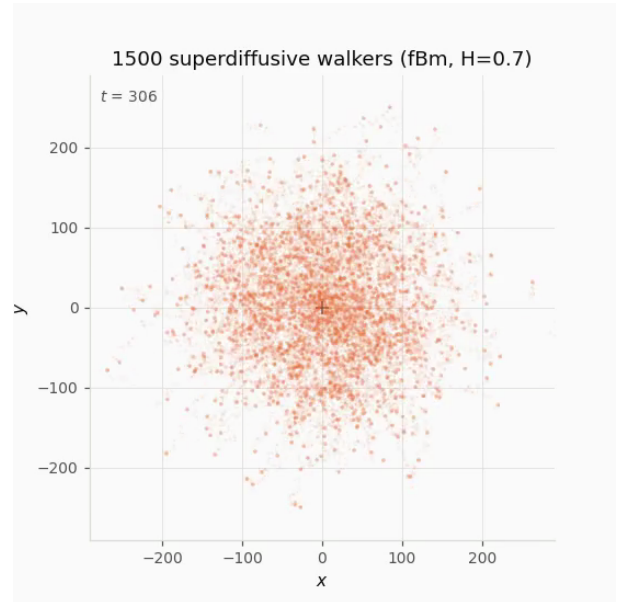


Figure 5: Increment autocorrelation $\rho(k)$ for five configurations, against the closed-form fBm prediction (dashed). Brownian motion and CTRW are indistinguishable at $\rho(k) = 0$ for $k \geq 1$ (delta-correlated increments); fBm at $H=0.3$ dips negative at lag 1 (antipersistence) while $H=0.7$ stays positive and decays slowly (persistence); the Lévy walk’s increments stay strongly positively correlated out to lag 10, the ballistic-stretch signature.



(a) Eight independent CTRW walkers, same parameters (frame from `results/media/ctrw_few.mp4`).



(b) 1500 superdiffusive walkers, fBm $H=0.7$ (frame from `results/media/fbm_cloud.mp4`).

Figure 6: Representative stills from the trajectory animations regenerated in this report’s re-run. The CTRW panel shows the qualitative signature of weak ergodicity breaking directly: walkers alternate between long flat pauses and short jumps, so different realizations of “the same process” visibly cover very different territory in the same elapsed time — the trajectory-level picture behind Table 3’s CTRW row.

3 Phase 2 — Estimator identifiability

3.1 Problem

Given a trajectory length T and an ensemble size N , what is the bias and variance of each estimator, per mechanism, and can mechanisms that share an exponent be told apart from a single trajectory at all? This is Q1 made quantitative and Q2 made operational.

3.2 Approach

Phase 2 builds 19 scale-invariant features (invariant under $x \mapsto cx$, verified for every feature on every mechanism) grouped into seven families (TA-MSD shape, increment ACF, non-Gaussian parameter, immobility, p -variation, excursion, non-stationarity; Table 5), trains pairwise and multi-class classifiers on them, compares a Whittle (spectral maximum likelihood) estimator against Phase 1’s naive TA-MSD fit relative to the Cramér–Rao bound, and closes the loop with a conditional exponent-correction pipeline: classify the mechanism, then apply a mechanism-specific bias correction before reporting $\hat{\alpha}$.

Table 5: Feature families used for mechanism classification, $T \in \{128, 512, 2048, 8192\}$, 120 trajectories per mechanism per length, 6 mechanisms (`results/phase2/step2_dataset_summary.json`).

Family	Representative features
<code>tamds</code>	TA-MSD fitted exponent, TA-MSD ratio
<code>acf</code>	increment autocorrelation at lags 1, 2, 3, 5, 10
<code>ngp</code>	non-Gaussian parameter at lags 1, 5, 10
<code>immobility</code>	immobile fraction, mean/max gap length
<code>p-variation</code>	p -variation ratios at $p=1, 2, 3$
<code>excursion</code>	maximal excursion ratio, range-to-std ratio
<code>nonstationarity</code>	increment variance ratio (early vs. late)

3.3 Bottleneck told in full: the Lévy-walk exponent correction

The conditional-correction pipeline initially fell through to Phase 1’s naive TA-MSD fit for the Lévy walk regardless of whether the mechanism was known or predicted — a review flagged that `mae_naive`, `mae_oracle` and `mae_predicted` were bit-identical for that one mechanism in `step6_conditional_exponent.json`. Three approaches were tried before one worked.

1. **GLS-weighted TA-MSD fit (failed).** PROJECT.md’s own fitting convention prescribes bootstrapping the lag–lag covariance of $\log(\text{TA-MSD})$ and weighting the log-log fit by its inverse. Tried four ways (full covariance with ridge regularization, diagonal-only inverse variance, rank-truncated pseudo-inverse, and a control using the *true* ensemble covariance from 800 independent trajectories). Every variant was worse than plain OLS; the true-covariance control was dramatically worse ($\text{MAE} \approx 0.43$ vs. naive’s ≈ 0.13). Diagnosis: a single trajectory’s short lags are precise but biased toward the ballistic within-flight regime, while its long lags are asymptotically correct but individually noisy. Inverse-variance weighting does exactly what it is built to do and concentrates weight on the precise-but-biased end, making the bias worse rather than better — a mismatch between the weighting objective (minimize variance) and the actual failure mode (lag-dependent bias), not a bootstrap-implementation bug.

2. **Moment-spectrum kink fit (failed).** The moment spectrum $\nu(q)$ is the Lévy walk’s textbook fingerprint, but resolving its high- q branch needs an ensemble, which one trajectory does not have. Direct single-trajectory fit: MAE ≈ 0.41 . Building a pseudo-ensemble by splitting one trajectory into 16–192 segments and sweeping the segment count and fit window did better but plateaued at MAE ≈ 0.147 – 0.15 , still worse than naive’s ≈ 0.13 .
3. **Flight-duration MLE via sub-grid reconstruction (adopted).** Both prior attempts reweighted or refit a statistic *derived* from the trajectory. Instead, mirroring the CTRW correction (which already works by doing MLE directly on waiting-time durations), flight durations are reconstructed from the gridded trajectory: flight speed is estimated as $\max(|\text{diff}|)$, each grid step is classified as “pure” (interior to one flight) or “straddling” (containing a turning point), and a straddling step’s fractional turning time is solved exactly from its one displacement value once neighbouring pure steps fix the flight directions. This sub-grid resolution matters because τ_0 is order-1, the same order as the grid spacing dt , so a large fraction of true flights are shorter than one grid step. A Hill estimator on the upper tail of the reconstructed durations gives a tail index $\hat{\gamma}$; the final exponent blends the naive fit and the correction, $w \hat{\alpha}_{\text{naive}} + (1-w)(3-\hat{\gamma})$, rather than trusting the Hill estimate alone (its reliability depends on the flight count, which shrinks as $\gamma \rightarrow 1$). A ceiling check — a Pareto MLE on the *true*, unquantized event durations — reaches MAE ≈ 0.05 against naive’s ≈ 0.13 , confirming the signal is real; the practical grid-reconstructed version recovers part of it.

Tuning pitfall. The first tuning pass fixed $\tau_0=\text{speed}=1$ and found $w=0.6$, $\text{top_frac}=0.4$ cut MAE by $\sim 20\%$, validated across 8 seeds. Re-running with the actual sampling distribution ($\tau_0, \text{speed} \sim U(0.5, 2.0)$) showed almost no improvement ($0.1161 \rightarrow 0.1163$): a varying τ_0 weights the harder-to-reconstruct short-flight regime more heavily than the fixed- τ_0 tuning set did, and the first-pass parameters had overfit to the easier case. Retuning against 600 realistically-drawn trajectories ($w=0.6$, $\text{top_frac}=0.35$) gave a held-out improvement of $0.133 \rightarrow 0.121$ ($\sim 9\%$). **Lesson: tune and validate against the actual parameter-generating distribution, not a convenient fixed-parameter subset of it** — one additional free parameter changed which correction looked best.

3.4 Final results

Table 6 shows multi-class classification accuracy (gradient boosting, the best of three classifiers tried) as a function of trajectory length, and Table 7 shows the three hardest-to-separate mechanism pairs individually. fBm vs. CTRW — the pair chosen specifically because they share an exponent but not an ergodic property — separates completely at every length tested; ablation confirms that `immobile_fraction` alone achieves 100% pairwise accuracy. SBM vs. `Brownian` separates only asymptotically, improving steadily as T grows. DDM vs. `Brownian` is already well separated by $T=128$ via the non-Gaussian parameter, which is the short-time signature of a fluctuating diffusivity.

Feature ablation at $T=2048$ (baseline accuracy 0.928) shows the non-Gaussian-parameter and non-stationarity families carry the most irreplaceable information: removing `ngp` drops accuracy to 0.839 and removing `nonstationarity` to 0.857, while removing any other single family costs less than one percentage point (Table 8). At the single-feature level, `immobile_fraction` alone reaches 100% accuracy on fBm-vs-CTRW and `ngp_lag1` alone reaches 100% on DDM-vs-Brownian — each pair has one feature that is, on its own, already sufficient.

Table 9 compares Phase 1’s naive TA-MSD fit against a Whittle (spectral) maximum-likelihood estimator, relative to the Cramér–Rao lower bound, for fBm at $H=0.3$ and $H=0.7$. The Whittle estimator is close to asymptotically efficient by $T=8192$; the naive estimator’s efficiency *falls* as T

Table 6: Six-way mechanism classification accuracy, gradient boosting classifier (`results/phase2/step3_classification_summary.json`). “Interp.” trains and tests on the same length; “extrap.” trains on a pooled set and tests on held-out trajectories of that length.

T	Interp. accuracy	Extrap. accuracy
128	0.747	0.786
512	0.861	0.913
2048	0.928	0.963
8192	0.961	0.950

Table 7: Pairwise classification error for the three hardest-to-separate mechanism pairs (`results/phase2/step3_classification_summary.json`, `step4_ablation_summary.json`).

Pair	$T=128$	$T=512$	$T=2048$	$T=8192$
fBm vs. CTRW	0.000	0.000	0.000	0.000
SBM vs. Brownian	0.296	0.208	0.100	0.050
DDM vs. Brownian	0.025	0.000	0.000	0.000

Table 8: Leave-one-family-out ablation, gradient boosting, $T=2048$ (`results/phase2/step4_ablation_summary.json`). Lower accuracy after removal means the family mattered more.

Configuration	Accuracy
Full feature set (baseline)	0.928
without <code>ngp</code>	0.839
without <code>nonstationarity</code>	0.857
without <code>acf</code>	0.922
without <code>immobility</code>	0.922
without <code>tamds</code>	0.929
without <code>excursion</code>	0.929
without <code>p_variation</code>	0.932

grows, showing its spread is a limitation of the estimator, not of the information present in the data.

Table 9: Estimator efficiency relative to the Cramér–Rao bound (results/phase2/step5.information.floor.json).

	Efficiency at $T =$			
	128	512	2048	8192
$H=0.3$, naive TA-MSD	0.448	0.320	0.227	0.094
$H=0.3$, Whittle MLE	0.779	0.813	0.743	0.968
$H=0.7$, naive TA-MSD	0.289	0.163	0.063	0.036
$H=0.7$, Whittle MLE	0.568	0.556	0.568	0.704

Finally, Table 10 closes the loop: naive Phase-1-style fitting, prediction-corrected fitting (classify, then apply the mechanism-specific correction, including the Lévy-walk correction above), and an oracle that already knows the true mechanism, at $T=2048$. The correction pipeline cuts overall MAE by 68.8% ($0.1687 \rightarrow 0.0527$), most of the way to the oracle’s 0.0478; the residual gap of 0.0049 is the cost of occasionally misclassifying the mechanism.

Table 10: Exponent recovery MAE by mechanism, $T=2048$ (results/phase2/step6.conditional.exponent.json).

Mechanism	Naive	Predicted	Oracle
Brownian	0.0532	0.0028	0.0000
fBm	0.0619	0.0272	0.0206
CTRW	0.3519	0.0906	0.0906
SBM	0.3654	0.0761	0.0627
DDM	0.0634	0.0064	0.0000
Lévy walk	0.1161	0.1132	0.1132
Overall	0.1687	0.0527	0.0478

No figures accompany Phase 2 in the current repository: its results are tables and JSON summaries rather than saved plots, so they are reported here as tables only, per this report’s sourcing rule.

4 Phase 3 — Mean-field coupling

4.1 Problem

Once agents are coupled through a mean field, how much does a single agent’s inferred regime tell you about the true collective regime, and does the field help (Q3)? This requires a coupled model with a well-defined “collective regime,” a validated way to measure both the agent-level and field-level exponent, and a comparison of observers that do and do not use the field.

4.2 Model

Each agent’s velocity follows the $\text{AR}(\infty)$ representation of an $\text{ARFIMA}(0, d, 0)$ process, truncated at K lags, with consensus coupling to the instantaneous mean field:

$$v_i(t+1) = \sum_{s=1}^K \kappa(s) v_i(t+1-s) + J(\langle v \rangle(t) - v_i(t)) + \sigma \eta_i(t+1), \quad \kappa(s) = \frac{\Gamma(s-d)}{\Gamma(-d)\Gamma(s+1)}.$$

The consensus form $J(\langle v \rangle - v_i)$, not an additive $+J\langle v \rangle$, is essential: because $\sum_s \kappa(s) = 1$, the additive form would make the collective mode unstable for any $J > 0$. With consensus coupling, the J term cancels exactly under all-to-all averaging, so the mean field $\langle v \rangle(t)$ is itself exactly $\text{ARFIMA}(0, d, 0)$ with variance σ^2/N , *for every value of J* — the **conservation law** this phase’s validation is built around. The shared asymptotic exponent is

$$\alpha = 1 + 2d, \quad d \in (-\tfrac{1}{2}, \tfrac{1}{2}).$$

The deviation from the mean field, $\delta_i = v_i - \langle v \rangle$, has a closed-form spectrum $S_\delta(\omega) = \sigma^2 / |(1 - e^{-i\omega})^d + J e^{-i\omega}|^2$, and its low-frequency behaviour depends on the **sign of d** :

- $d > 0$ (superdiffusive memory): the denominator tends to J as $\omega \rightarrow 0$, so $S_\delta(0) = \sigma^2/J^2$ is finite — the deviation becomes ordinary diffusion at long times, with memory surviving only below a crossover $\tau_c \sim J^{-1/|d|}$.
- $d < 0$ (subdiffusive memory): the fractional term diverges as $\omega \rightarrow 0$ and dominates J at every reachable frequency, $S_\delta(\omega) \sim \omega^{2|d|} \rightarrow 0$. Coupling does *not* destroy the anomalous exponent, and τ_c has no meaning.

This is the section’s central physical result, stated plainly: **coupling kills superdiffusive memory but not subdiffusive memory**. Whether an isolated single-agent observer can still reach the true collective regime depends on which side of zero d is on.

4.3 Bottlenecks, told as a sequence

This phase had the most iterations of any in the project; each one changed what could legitimately be measured next.

V1: wrong reference process.

The original specification claimed the $J=0$ limit was exactly fractional Gaussian noise (fGn). Validating against fGn’s closed-form autocorrelation showed a persistent gap (~ 0.08 – 0.09 at lag 1) that did *not* shrink between $K=64$ and $K=1024$, ruling out ordinary truncation error. The correct reference is $\text{ARFIMA}(0, d, 0)$, not fGn: the two share the asymptotic exponent

$\alpha=1+2d$ but not the short-lag correlation ($\rho(1) = d/(1-d)$ for ARFIMA vs. $2^{2d} - 1$ for fGn). Re-validating against the corrected target dropped the maximum deviation from ~ 0.08 to 0.005–0.007 (Table 11) — the simulator was correct all along; the specification’s reference process was not.

V2/V3: sweep recalibration and a normalization bug.

The default J grid predicted τ_c values spanning 16 to 160,000 at $d=0.25$, only one of which fell inside the trustworthy $K/10$ window; the grid was replaced with one targeted directly at $\tau_c \in \{8, 16, 32, 64\}$. The deviation-spectrum validation (V3) then showed the simulator and the closed-form spectrum agreeing in shape but differing by a flat factor of 2π — a missing normalization constant in the spectral-density formula, fixed, after which the mean absolute log-ratio in the trustworthy band fell to 0.036–0.037 (Table 12).

V4: redesigned from a crossover measurement to a joint fit.

The first attempt to locate τ_c from the local slope of an agent’s own MSD gave a slope of $\log \tau_c$ vs. $\log J$ of $+1.09$ against a predicted -4.0 — wrong sign, and not fixable by averaging more realizations. The diagnosis: the recalibrated J grid makes J comparable to or larger than $\kappa(1) = d$, so the local-slope measurement was probing a regime the low-frequency crossover argument was never actually valid for — it is not that the measurement failed, it is that nothing predicted it should succeed. V4 was redesigned as a joint (d, J) Whittle MLE against the deviation spectrum instead, which fits the whole spectral shape rather than one feature of it (Table 13). An optimizer bug was found en route: a single L-BFGS-B run from the bounds’ midpoint converged to a poor local optimum whenever the true J was far from that midpoint (most of the grid); fixed with a 3×3 multi-start grid. The original crossover measurement is kept only as a qualitative illustration (Figure 7), explicitly not asserted as a measurement.

Step 2: a naive, coupling-blind Whittle fit is misleading for $d < 0$.

Fitting the correct, coupling-aware two-parameter model recovers the true d regardless of J by construction — informative for Q3’s model-aware observer, but useless for *demonstrating* the phenomenological crossover this step exists to show. A coupling-blind, one-parameter Whittle fit shows the expected decline for $d > 0$, but *also* declines sharply for $d < 0$ (contradicting the “subdiffusive memory survives” prediction): the $+Je^{-i\omega}$ term reshapes the whole spectrum, not just its low-frequency limit, so forcing a one-parameter model onto genuinely two-parameter data biases the fit regardless of the true sign of d . Resolved by using a model-free, windowed ensemble-averaged MSD fit as the primary “typical-agent” measure instead — it cannot inherit a spectral model’s misspecification because it assumes no spectral model at all.

Close-out: two field-side Whittle bugs, found late, that silently changed earlier numbers.

Checking V5’s variance ratio (expected 1.00 from the analytic Fisher information of ARFIMA’s d , measured 3.6–3.9) surfaced two separate bugs: (1) the new ARFIMA Whittle fits never profiled out the noise scale σ^2 , which is harmless for an individual agent (true scale exactly 1) but catastrophic for the mean field (true scale σ^2/N), pinning $\hat{d}$ at the search bound; (2) the function used for *every* “coherent”/“field” number in the phase so far (V2, step 2’s coherent regime, step 3’s field-aware observer, step 4’s field correlations) was fitting **fGn, not ARFIMA** to the field — the identical mistake V1 had already found and fixed for a single agent, never caught on the field side because nothing had checked the field’s fitted variance against its own analytic efficiency until V5 did. Both fixed; every field-side number in this report reflects the corrected estimator, and the report states explicitly, below, which earlier-reported numbers this superseded.

Item 3: a genuine optimizer bug, not just a sampling-noise story, for $d < 0$.

A residual-surface diagnostic (a full 2-D grid search over (d, J) , not just a 1-D scan) found that for $d=0.25$, boundary-pinned fits are a real finite-sample identifiability wall (a dense 7×7 restart grid does not move them). For $d=-0.25$, the same diagnostic found the multi-start grid landing in the *wrong basin* for roughly a third of agents — a real, fixable optimizer bug, distinct from the $d > 0$ case. Fixed by widening `whittle_dj_estimate`’s multi-start grid from 3×3 to 7×7 starts, re-targeted to cover negative- d , large- J corners the old grid could not reach (Table 15). This did not change the phase’s headline sign-asymmetry finding, which was established via the model-free windowed MSD fit and never touched by the bug — but it materially improved the model-aware observer’s reported bias for $d < 0$, strong- J populations in steps 3 and 4.

Table 11: V1: ARFIMA reference-process validation, $N=800$, $T=512$, $K=1024$ (`results/phase3/step1_v1_reduction.json`).

d	max ACF dev.	EA-MSD fitted	exact ARFIMA target	asymptote $1+2d$
-0.25	0.0053	0.571	0.543	0.5
0.00	0.0049	1.033	1.000	1.0
0.25	0.0074	1.513	1.495	1.5

Table 12: V3: deviation-spectrum validation after the 2π normalization fix, $J=0.3$, $N=500$, $K=1024$ (`results/phase3/step1_v3_deviation_spectrum.json`).

d	Mean abs. log-ratio (trustworthy band)	n frequencies
0.25	0.0365	502
-0.25	0.0365	502

Table 13: V4: joint (d, J) Whittle recovery, $N=400$, $T=1024$, $K=1024$, $J < |d|/2$ grid (`results/phase3/step1_v4_whittle_dj.json`).

d	mean $\hat{d}$	max $ \hat{d} - d $	std($\hat{d}$) across J
0.15	0.1501	0.0022	0.0014
0.25	0.2493	0.0027	0.0023
0.35	0.3453	0.0084	0.0056
0.45	0.4467	0.0073	0.0058

4.4 Final results

Sign-asymmetry confirmation (step 2). Table 14 reports the windowed ensemble-averaged MSD exponent for the “typical agent” at the weakest and strongest coupling tested, for all eight d values swept. For every $d > 0$, the typical-agent exponent falls substantially toward 1 as J grows; for every $d < 0$, it stays within about 0.005–0.05 of its $J=0$ value. The coherent (centre-of-mass) exponent is flat across J for every d , confirming the conservation law directly in the estimator rather than only in the field’s own spectrum.

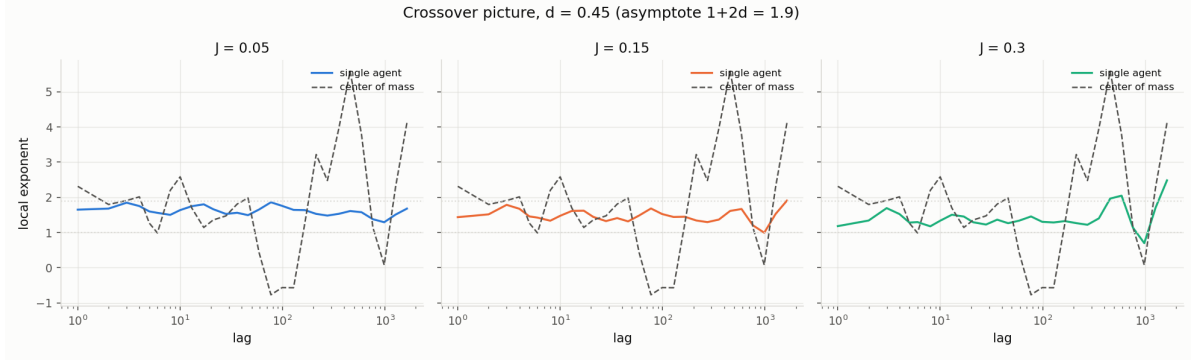


Figure 7: Local MSD exponent vs. lag for a single agent (colour) and the centre-of-mass mode (dashed), $d=0.45$, $K=16384$, three coupling strengths. Kept explicitly as a qualitative illustration, not a measurement (Section 4 V4 discussion): the single-agent curve stays well below the coherent asymptote $1+2d=1.9$ throughout, while the centre-of-mass curve (one realization, hence visibly noisy) swings widely because it is a single trajectory’s local slope, not an ensemble average.

Table 14: Typical-agent vs. coherent exponent across the coupling sweep, $N=400$, $T=1024$, $K=1024$ (`results/phase3/step2_two_regimes.json`). $J_{\max}$ differs by d group (see source file); it is chosen so τ_c stays inside $K/10$.

d	asymptote $1+2d$	$\alpha_{\text{typical}}(J=0)$	$\alpha_{\text{typical}}(J_{\max})$	α_{coherent}
-0.45	0.10	0.252	0.257	0.103
-0.35	0.30	0.387	0.383	0.270
-0.25	0.50	0.551	0.517	0.457
-0.15	0.70	0.733	0.626	0.655
0.15	1.30	1.338	1.064	1.260
0.25	1.50	1.541	1.182	1.461
0.35	1.70	1.722	1.269	1.663
0.45	1.90	1.858	1.337	1.866

Observer bias/variance (step 3). Four observers are compared: isolated model-agnostic (Phase 2’s classifier+correction pipeline on the agent’s own position), isolated model-aware (joint Whittle fit on the agent’s own velocity), field-aware (Whittle fit on the mean field), and a simple average of model-aware and field-aware. Figure 8 shows the full bias/variance curves; the qualitative picture is mechanism-independent of the sign of d : the model-agnostic observer’s bias grows sharply and monotonically with J in the same direction regardless of the sign of d (consistent with step 2’s finding that naive estimators are pulled toward antipersistence by coupling); the model-aware observer stays nearly unbiased but its variance grows with J and is the largest of the four at every point; the field-aware observer’s bias and variance are *exactly* flat across J , down to the last reported digit — the conservation law again, now visible in an estimator rather than in the field’s own spectrum; the combination sits between model-aware and field-aware, closer to field.

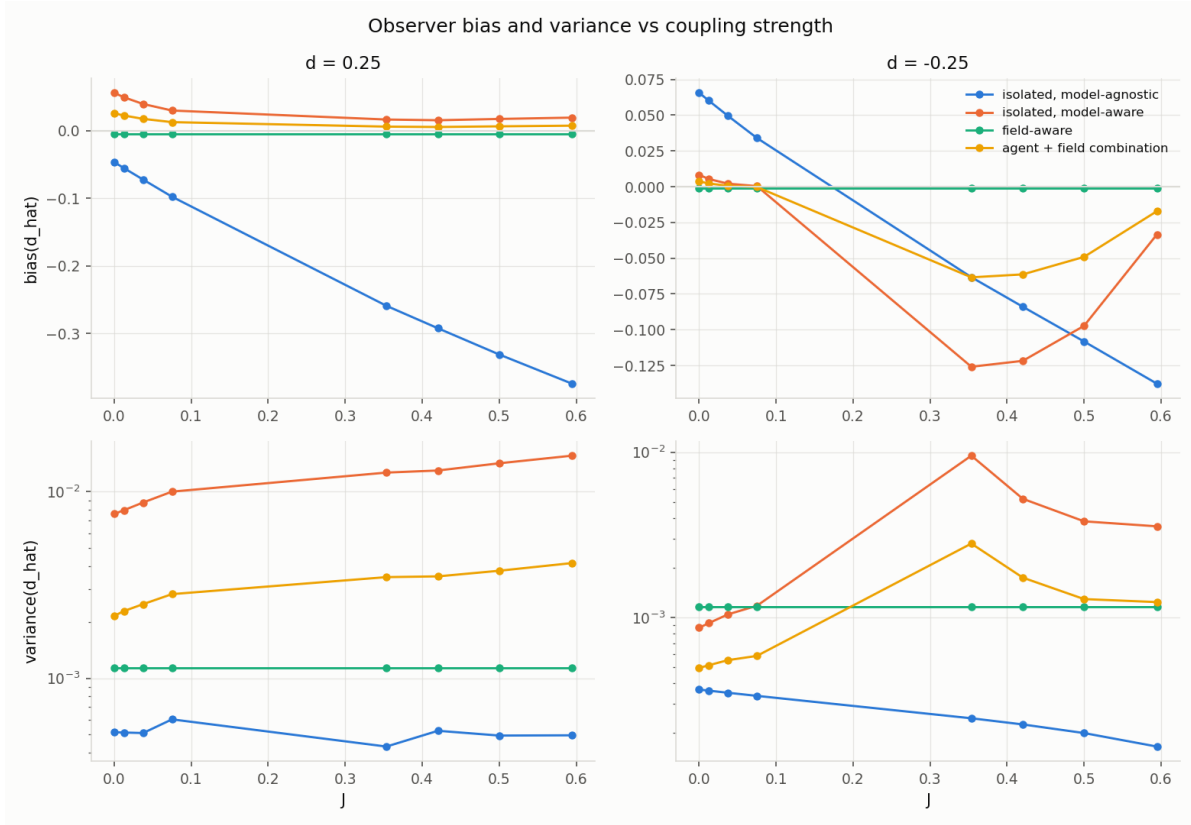


Figure 8: Bias (top) and variance (bottom, log scale) of $\hat{d}$ vs. coupling strength J , four observers, both signs of $d = \pm 0.25$. The field-aware curve (green) is a flat line in every panel; the model-agnostic curve (blue) is the only one whose bias keeps growing without bound as J increases.

Residual-surface diagnostic (item 3 close-out). A full 2-D grid search over (d, J) separates the two candidate explanations for the model-aware observer’s poor performance at strong coupling (Table 15): for $d = 0.25$, boundary-pinned fits are a genuine finite-sample wall (unchanged by a denser search grid); for $d = -0.25$, roughly a third of the pinned fits were the optimizer landing in the wrong basin, fixed by widening the multi-start grid, which roughly halved the observer’s bias at strong coupling on that side.

Table 15: Residual-surface diagnostic, 60 sampled agents per point, post-fix multi-start grid (`results/phase3/step3_residual_surface.json`).

Point	Pinned	Bias (incl. pinned)	Bias (excl. pinned)	Grid matches optimizer
$d=0.25, J=0.595$	10/60	0.020	-0.024	yes
$d=-0.25, J=0.354$	6/60	-0.070	-0.051	yes

Q3 correlation (step 4). With $d \sim U(-0.4, 0.4)$ drawn fresh per trial, raw correlation between an observer’s $\hat{d}$ and the true coherent d is uninformatively high for all four observers (0.91–0.99 throughout); normalized RMSE and R^2 separate them clearly instead (Table 16, Figure 9). Field-aware is exactly flat across J ; model-agnostic degrades the worst, reaching a *negative* R^2 at $J=0.5$ (worse than predicting the mean every time).

Table 16: Q3 metric by observer and coupling strength, NRMSE / R^2 , $d \sim U(-0.4, 0.4)$, $N=400$, $T=1024$, $K=1024$ (`results/phase3/step4_q3_correlation.json`).

J	Model-agnostic	Model-aware	Field-aware	Combination
0.0	0.301 / 0.909	0.268 / 0.928	0.135 / 0.981	0.160 / 0.974
0.1	0.376 / 0.858	0.330 / 0.890	0.135 / 0.981	0.185 / 0.966
0.3	0.757 / 0.424	0.536 / 0.711	0.135 / 0.981	0.277 / 0.923
0.5	1.131 / -0.289	0.529 / 0.718	0.135 / 0.981	0.285 / 0.918

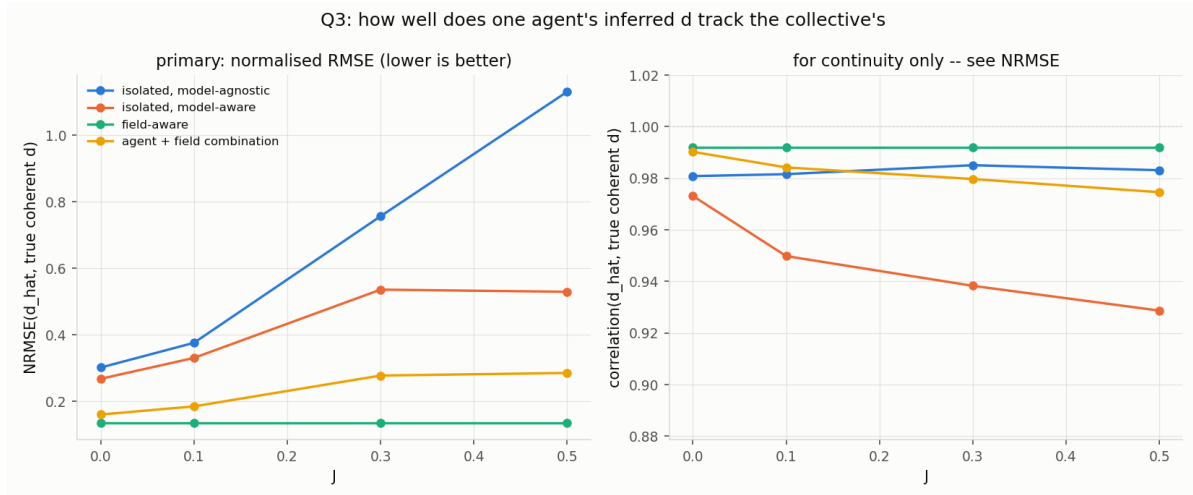


Figure 9: Left: normalized RMSE of $\hat{d}$ against true coherent d , the primary Q3 metric — lower is better. Right: raw correlation, kept for continuity but nearly flat and uninformative across observers; the left panel is where the four observers actually separate.

Scaling with N and K . Table 17 shows an unexplained but statistically real ($p=0.046$ on an F-test between the extremes) rise in field-aware estimator variance with N ($\text{var} \sim N^{0.27}$), contradicting the flat prediction the conservation law and V5 both establish at $N=400$ — boundary pinning was checked and ruled out as the cause; the true mechanism is unresolved and flagged for Phase 4.

Table 18 shows wall-clock cost vs. memory length K , measured uncontended on the machine this report was produced on: the per-step cost is $O(NK)$ and the burn-in itself scales as $O(K)$, giving a baseline slope of 2 on a log-log plot. The first two $4\times$ steps in K cost $10.6\times$ and $17.5\times$ respectively (below and close to the quadratic prediction of $16\times$), but the step from $K=4096$ to $K=16384$ costs $208.9\times$ more — thirteen times above the quadratic baseline — pulling the overall fitted slope to 2.49. This is the same qualitative anomaly NOTES.md reports from its own reference machine (a sharp, localized cost jump once K exceeds a few thousand, consistent with the (N, K) memory ring buffer outgrowing cache), but larger in magnitude here; absolute wall-clock numbers in a timing study are hardware-specific and are not expected to match across machines, only the qualitative shape.

Table 17: Estimator variance vs. population size N , $d=0.25$, $T=1024$, $K=1024$ (`results/phase3/step4_n_scaling.json`).

N	Field var. ($J=0$)	Field var. ($J=0.3$)	Isolated var. ($J=0$)	Typical-agent MSD var. ($J=0.3$)
50	0.000248	0.000248	0.00249	0.01555
100	0.000340	0.000340	0.00249	0.00497
400	0.000417	0.000417	0.00249	0.00113
1600	0.000555	0.000555	0.00249	0.00042

A more heavily sampled follow-up at $J=0$ only (20 realizations instead of 10, `step4_field_variance_vs_n.json`) confirms the same monotonic rise with tighter confidence intervals: variance $0.000383 \rightarrow 0.000544 \rightarrow 0.000812 \rightarrow 0.000986$ across the same four N values, a ratio of $2.57\times$ between $N=1600$ and $N=50$ (up from $2.24\times$ at half the realizations) — the effect strengthens under more data, not a statistical fluke.

Table 18: Wall-clock cost vs. memory length K , $N=50$, $T=64$, burn-in = $2K$, single uncontended run, regenerated for this report (`results/phase3/step4_k_scaling.json`).

K	Burn-in	Elapsed (s)	Ratio over prior K ($4\times$ increase)
256	512	0.026	—
1024	2048	0.272	$10.6\times$
4096	8192	4.759	$17.5\times$
16384	32768	994.111	$208.9\times$

Fitted log-log slope: $p = 2.49$.

V5: is the field really N -fold more precise (close-out gate)? Table 19 is the corrected version of the comparison that motivated the close-out bug hunt above: at $J=0$, where both an isolated agent and the N -agent mean field can legitimately use the same one-parameter model, Whittle precision for a spectral *shape* parameter does not depend on series amplitude, so the two variances should be comparable, not N -fold apart. After both field-side bugs were fixed, the ratio sits within $\sim 10\%$ of the analytic prediction of 1.00, and both arms sit within $\sim 10\%$ of the Cramér–Rao bound $6/(T\pi^2) = 5.94 \times 10^{-4}$. This was the supervisor’s explicit gate for Phase 4 and it is cleared.

Table 19: V5 corrected efficiency check, $N=400$, $T=1024$, $K=1024$, 40 realizations $\times$ 20 agents, $J=0$ (`results/phase3/step3_v5_check.json`, regenerated for this report).

d	Isolated efficiency	Field efficiency	Ratio (isolated/field)
0.25	0.987	1.092	1.106
-0.25	0.987	1.074	1.088

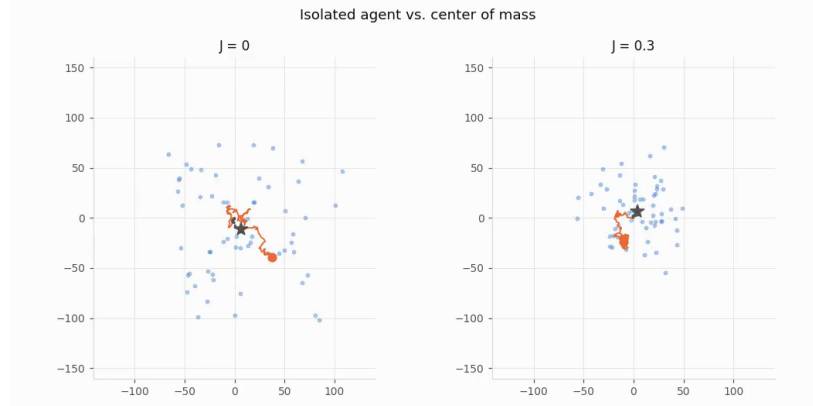


Figure 10: Frame from the mean-field comparison animation (`results/media/phase3_figures/fig4_meanfield_animation.mp4`), $N=60$, $T=400$, $K=512$: an isolated agent's trajectory (orange) and the population's centre of mass (star) against the full population (faint points). The animation, not reproducible in a static panel, shows the isolated agent's excursions being visibly damped relative to the spread of the population as coupling increases from $J=0$ to $J=0.3$.

5 Overall conclusion

Q1 (single-trajectory estimator reliability) is answered concretely, not just qualitatively: a single trajectory is unreliable in two distinct ways depending on the mechanism, and the two look different from the inside. Scaled Brownian motion converges tightly onto a wrong exponent, so the estimator’s own precision gives no warning; CTRW converges slowly onto a wrong exponent and keeps a wide, barely-shrinking spread, so at least the estimator is honest about its own uncertainty. Both report $\hat{\alpha} \approx 1$ from a single trajectory for unrelated reasons, and that coincidence is the concrete obstacle Phase 2 exists to remove.

Q2 (mechanism discrimination) is largely solved within the scope tested: nineteen scale-invariant features, mostly drawn from the propagator shape and the immobility structure rather than the exponent itself, separate every mechanism pair tested, several of them (fBm vs. CTRW, DDM vs. Brownian) almost perfectly even at short trajectory lengths. Once the mechanism is known, a mechanism-specific correction — including one built from first principles for the Lévy walk, after two more natural approaches failed for diagnosable reasons — closes most of the gap to an oracle that already knows the mechanism.

Q3 (single agent vs. collective) has a genuinely asymmetric answer, and that asymmetry is the phase’s central physical finding: coupling destroys a superdiffusive agent’s ability to reveal the collective regime from its own short-term behaviour, but does not destroy a subdiffusive agent’s ability to do so. The field itself is always informative and its precision is provably independent of the coupling strength (the conservation law, confirmed at the level of an estimator’s bias and variance, not only in the field’s own spectrum) — so a field-aware observer is the most reliable of the four tested at every coupling strength, and a naive, coupling-blind observer is actively misleading, not merely noisier, once coupling is non-trivial.

Q4 (modular heterogeneity) remains open: Phase 4 has not been run. What Phase 3 establishes as its precondition is concrete rather than aspirational: a validated coupled-agent simulator, an estimator toolbox whose efficiency is measured (not assumed) against the relevant Cramér–Rao bounds at the scale it was tested ($N=400$), and an explicit list of what is *not* yet known at other scales — field-aware variance should not be assumed N -independent when budgeting Phase 4 (Table 17 shows a real, unexplained rise with N), and wall-clock cost is super-linear in both N and K with a sharp, localized cost jump once K exceeds a few thousand (Table 18). These are the concrete numbers Phase 4’s design should be budgeted against, not a smooth extrapolation from the $N=400$ scale this report’s results were measured at.